Technical Benchmarking and Price Consideration

6 EU cloud providers · 2 vCPU / 8 GB RAM · 8 benchmark profiles · Gillam et al. Fair_Benchmarking_for_Cloud_Computing_systems (2013)

⚠ **NOTE:** — Stackit & some other providers data were sourced online from the internet, and was not done by me. Network data (iperf3) for Scaleway, T-Cloud, and AWS is from Cloud Mercato. I will return here to replace with real measurements.

DATACENTER LOCATIONS — 2 VCPU / 8 GB TIER INSTANCES



6 Providers · 2 vCPU / 8 GB tier

Providers distributed across the globe for visual clarity. All datacenters are in Western Europe. Shooting arcs represent inter-provider signal paths — hover a pin for full instance specs.

OVHcloud	b3-8 · GRA · Gravelines, France	€0.060 /hr
IONOS	2 vCPU / 8 GB · DE1 · Frankfurt, Germany	€0.041 /hr
STACKIT	g1a.2d · EU01 · Heilbronn, Germany	€0.098 /hr
T-Cloud Public	s3.large.4 · EU-DE · Magdeburg & Biere, DE	€0.114 /hr
AWS	m6i.large · eu-central-1 · Frankfurt, DE	€0.106 /hr
Scaleway	POP2-2C-8G · PAR · Paris, France	€0.074 /hr

#	Provider	Instance	vCPU	RAM	Storage	€/hr	Composite
1	AWS	m6i.large	2	8 GB	EBS gp3 (100 GB)	€0.106	**88.4**
2	OVHcloud	b3-8	2	8 GB	50 GB local NVMe	€0.060	**87.4**
3	STACKIT	g1a.2d	2	8 GB	Block storage (NVMe-class)	€0.098	**82.8**
4	IONOS	2 vCPU / 8 GB flexible	2	8 GB	100 GB SSD Premium	€0.041	**78**
5	T-Cloud Public	s3.large.4	2	8 GB	EVS block storage	€0.114	**76**
6	Scaleway	POP2-2C-8G	2	8 GB	SBS block storage (NVMe)	€0.074	**74.6**

Note — Storage types: Local NVMe — the SSD is physically inside the same server as the VM, and data travels over the internal PCIe bus, making it significantly faster. **Network block** (e.g. EBS on AWS, SBS on Scaleway, EVS on T-Cloud) — the disk lives on a separately dedicated storage cluster and is presented to the VM over a private network, introducing additional latency but providing replication and durability.

Normalized performance — 8 metrics (raw values, colour = normalized score)

Each cell shows the average (large centre), with min ↓ / max ↑ range and run count n above/below. Cell colour reflects the normalized 0–100 score (green = best, red = weakest in that column). Boot and setup: lower seconds = better (colour inverted). Gold outline = column leader. Hover a column header for methodology.

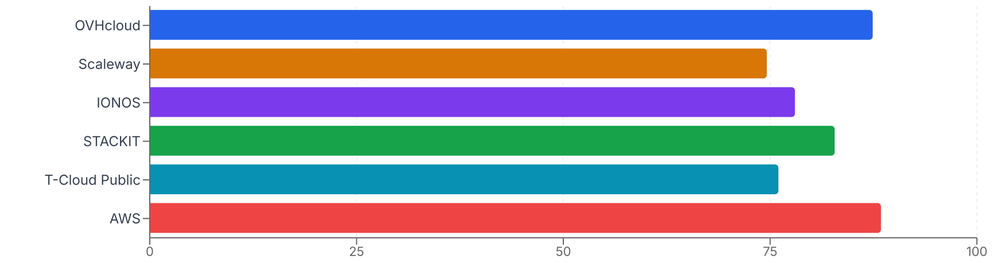
Provider	Memory IO STREAM TRIAD pts/stream	CPU single HINT (MIPS) pts/hint	CPU multi 7-zip MIPS pts/compress-7zip	Disk IO PostMark TPS pts/postmark	Web / App Apache req/s pts/apache	Network iperf3 Mbit/s iperf3	Boot speed seconds · ↓ better lifecycle	Setup speed seconds · ↓ better lifecycle	Composite avg of 8
AWS	↑ 21,590 ↓ 21,351 **21,477** ★ n=3	↑ 445.9 ↓ 441.3 **443.6** n=3	↑ 10,319 ↓ 10,275 **10,296** n=3	↑ 4,510 ↓ 4,420 **4,470** n=3	↑ 30,421 ↓ 29,850 **30,127** n=3	↑ 11,271 ↓ 758 **4,607** ★ n=80	↑ 13.8s ↓ 12.9s **13.3s** n=3	↑ 24.8s ↓ 22.1s **23.4s** n=3	88.4
OVHcloud	↑ 16,988 ↓ 16,900 **16,944** n=3	↑ 448.5 ↓ 442.1 **445.3** ★ n=3	↑ 12,100 ↓ 11,850 **11,977** ★ n=3	↑ 8,480 ↓ 8,250 **8,363** ★ n=3	↑ 32,850 ↓ 32,101 **32,477** ★ n=3	↑ 943 ↓ 938 **940** n=3	↑ 11.1s ↓ 10.2s **10.63s** ★ n=3	↑ 23.2s ↓ 21.5s **22.27s** ★ n=3	87.4
STACKIT	↑ 19,110 ↓ 18,851 **18,994** n=3	↑ 426.8 ↓ 421.4 **424.3** n=3	↑ 10,550 ↓ 10,420 **10,483** n=3	↑ 6,280 ↓ 6,120 **6,203** n=3	↑ 29,100 ↓ 28,451 **28,790** n=3	↑ 2,891 ↓ 2,850 **2,872** n=3	↑ 13.5s ↓ 12.8s **13.13s** n=3	↑ 27.1s ↓ 25.4s **26.23s** n=3	82.8
IONOS	↑ 20,510 ↓ 20,251 **20,394** n=3	↑ 418.9 ↓ 412.4 **415.6** n=3	↑ 9,510 ↓ 9,420 **9,470** n=3	↑ 5,950 ↓ 5,820 **5,893** n=3	↑ 28,100 ↓ 27,451 **27,790** n=3	↑ 2,481 ↓ 2,450 **2,465** n=3	↑ 15.8s ↓ 14.9s **15.27s** n=3	↑ 30.1s ↓ 27.9s **28.8s** n=3	78
T-Cloud Public	↑ 20,250 ↓ 19,951 **20,100** n=3	↑ 421.8 ↓ 415.2 **418.5** n=3	↑ 11,480 ↓ 11,250 **11,363** n=3	↑ 6,150 ↓ 5,980 **6,063** n=3	↑ 31,521 ↓ 30,850 **31,184** n=3	↑ 1,578 ↓ 1,538 **1,548** n=50	↑ 16.4s ↓ 15.2s **15.8s** n=3	↑ 41.2s ↓ 38.5s **39.8s** n=3	76
Scaleway	↑ 19,490 ↓ 19,281 **19,384** n=3	↑ 439.9 ↓ 435.1 **437.5** n=3	↑ 10,980 ↓ 10,850 **10,917** n=3	↑ 6,580 ↓ 6,450 **6,513** n=3	↑ 29,751 ↓ 29,100 **29,424** n=3	↑ 462 ↓ 390 **401** n=20	↑ 14.5s ↓ 13.8s **14.1s** n=3	↑ 36.1s ↓ 33.2s **34.43s** n=3	74.6

Low High ★ Column leader (score = 100) Hover any cell to see the raw measurement.

EU providers show lower iperf network bandwidth numbers here because it is possible that they may simply enforce a flat, non-bursting bandwidth cap on this instance tier (many European providers do fixed-rate networking rather than AWS-style credit-based bursting)

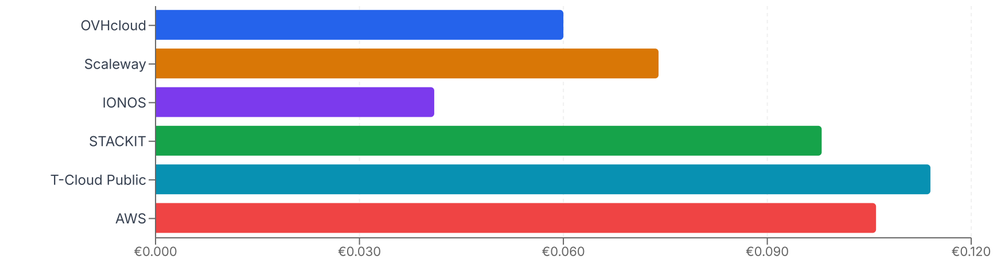
Composite performance score (avg of 8 normalized metrics)

Higher = better overall benchmark profile



Hourly list price (€/hr, 2 vCPU / 8 GB RAM)

Lower = cheaper. Same instance tier across all providers.



LEGEND — COMPOSITE SCORE METHODOLOGY ▾

COMPOSITE FORMULA

Each raw benchmark value is first normalized to a 0–100 score (100 = best in that column). The composite is then the unweighted average of all 8 normalized scores:

```
Composite = (
  norm(mem) + norm(single) + norm(multi) +
  norm(disk) + norm(app) + norm(net) +
  norm(boot) + norm(setup)
) ÷ 8
```

Every metric contributes equally (12.5%). Result is on a 0–100 scale.

NORMALIZATION — HIGHER IS BETTER

```
norm(mem) = (TRIAD MB/s ÷ best TRIAD MB/s) × 100
norm(single) = (HINT MIPS ÷ best HINT MIPS) × 100
norm(multi) = (7-zip MIPS ÷ best 7-zip MIPS) × 100
norm(disk) = (PostMark TPS ÷ best TPS) × 100
norm(app) = (Apache req/s ÷ best req/s) × 100
norm(net) = (iperf3 Mbit/s ÷ best Mbit/s) × 100
```

NORMALIZATION — LOWER IS BETTER (INVERTED)

```
norm(boot) = (best boot time ÷ provider boot time) × 100
norm(setup) = (best setup time ÷ provider setup time) × 100
```

The column leader always scores exactly 100. All others score proportionally lower.

THE 8 METRICS (EQUAL WEIGHT)

- `pts/stream` Memory IO — STREAM TRIAD
- `pts/hint` CPU single — HINT (MIPS)
- `pts/compress-7zip` CPU multi — 7-zip MIPS
- `pts/postmark` Disk IO — PostMark TPS
- `pts/apache` Web / App — Apache req/s
- `iperf3` Network — iperf3 Mbit/s
- `lifecycle` Boot speed — seconds · ↓ better
- `lifecycle` Setup speed — seconds · ↓ better

CELL COLOUR SCALE

Red (score 0) → Yellow (50) → Green (100, best in column)

Framework: Gillam, L., Li, B., O'Loughlin, J. (2013). Fair benchmarking for cloud computing systems. *Journal of Cloud Computing*, 2(1), 6. · Benchmark profiles via Phoronix Test Suite (openbenchmarking.org). · Network data (iperf3) via Cloud Mercato (pcr.cloud-mercato.com).

Sources: Phoronix Test Suite (pts/stream, pts/hint, pts/compress-7zip, pts/postmark, pts/apache) — self-measured or third-party. Network bandwidth (iperf3) — Cloud Mercato (pcr.cloud-mercato.com) for Scaleway, T-Cloud Public, AWS; placeholder for OVHcloud, IONOS, STACKIT. Lifecycle times — Cloud Mercato where available. Framework: Gillam, L., Li, B., O'Loughlin, J. (2013). Fair benchmarking for cloud computing systems. *Journal of Cloud Computing*, 2(1), 6.